

Advanced NLP Exercise 1: Open Questions & Qualitative Analysis

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Part 1: Open Questions (Section 1)

Question 1: QA Datasets for Intrinsic Concepts

Question Answering is often used as a framework to test core linguistic structures (intrinsic concepts) rather than just factual knowledge. Here are three examples:

1. QA-SRL (Semantic Role Labeling via Question Answering)

- **Explanation:** This dataset annotates predicate-argument structures by asking questions like "Who did what to whom?" based on verbs in a sentence. It measures an intrinsic property because it evaluates the model's grasp of core semantic roles and sentence structure, independent of external world knowledge.

2. QAMR (Question-Answer Meaning Representation)

- **Explanation:** Expanding on QA-SRL, QAMR uses QA pairs to capture the relationships between *all* words in a sentence, including nouns and adjectives (e.g., "What kind of car?"). It intrinsically measures whether a model understands the complete, granular semantic dependency graph of a sentence.

3. QA-Coref (Coreference Resolution via QA) / CorefQA

- **Explanation:** This method frames coreference resolution as a reading comprehension task by asking questions to link pronouns or noun phrases to their original entities (e.g., "Who does 'she' refer to?"). It measures the intrinsic linguistic ability to track entities and maintain grammatical coherence across a longer text.

Question 2: Inference-Time Scaling

(a) Methods for Inference-Time Scaling:

1. Self-Consistency

- **Description:** A technique to increase test-time compute by sampling a diverse set of reasoning paths instead of relying on a single greedy decode. The final answers are then aggregated by marginalizing out the individual reasoning paths.
- **Advantages:** Provides a more robust final answer by relying on multiple diverse logical approaches rather than a single attempt.
- **Computational Bottlenecks:** Generating multiple complete reasoning paths heavily consumes computational power (FLOPs).
- **Parallelizable?** Yes, each diverse reasoning path can be generated independently and simultaneously.

2. Automatic Verifiers

- **Description:** Utilizing automatic verifiers at test time to check the outputs of the language model. This can involve trained models or deterministic checks like regular expressions for math, unit-tests for code, or test questions for science.
- **Advantages:** Improves Chain-of-Thought generation by actively catching errors and verifying the validity of the steps or the final answer.
- **Computational Bottlenecks:** Requires additional compute to execute the verification steps (e.g., running code environments or evaluating a secondary reward model).
- **Parallelizable?** Yes, multiple candidate answers can be scored or verified in parallel.

3. Iterative Reasoning (Planning, Backtracking, and Self-evaluation)

- **Description:** A method where the length of the chain-of-thought is increased as a measure of inference scaling. The model learns to break down tricky steps (planning), evaluate its intermediate thoughts (self-evaluation), and try a different approach when the current one isn't working (backtracking).
- **Advantages:** Dramatically improves the model's ability to reason, allowing it to recognize and correct its own mistakes mid-generation.
- **Computational Bottlenecks:** Heavily consumes memory (VRAM) to maintain the extended context window of intermediate "thinking" tokens, and compute time for the long generation process.
- **Parallelizable?** No, the process is highly sequential as each reasoning step or backtrack depends entirely on the self-evaluation of the previous step.

(b) Scenario: Single GPU with Large Memory for Complex Scientific Reasoning

For a complex scientific task on a single GPU with large memory capacity, I would choose **Iterative Reasoning (Planning, Backtracking, and Self-evaluation)**.

Why? Complex scientific reasoning often requires breaking tricky problems into simpler steps and recognizing mistakes mid-process. Because you only have a single GPU, heavily parallelized methods like Self-Consistency would have to run sequentially, causing a massive bottleneck in time. However, because your GPU has a *large memory capacity*, it is perfectly suited to handle the extended context window and long chain-of-thought required for a model to backtrack and evaluate its own intermediate thoughts. This allows deep, deliberate problem-solving without running out of VRAM. Alternatively, an **Automatic Verifier** tailored to "test questions for science" could be combined with this approach to validate the final output once the reasoning chain concludes.

Part 2: Qualitative Analysis (Section 2.1, part 2)

Did the configuration that achieved the best validation accuracy also achieve the best test accuracy?

No. The configuration that achieved the highest validation accuracy (Run 2: 3 epochs, lr=3e-5) achieved a test accuracy of 83.25%. Surprisingly, the model with the lowest validation accuracy (Run 3: 5 epochs, lr=3e-5) actually achieved the highest test accuracy at 84.00%.

Analysis of Best (3-Epoch) vs. Worst (5-Epoch) Configurations:

The configuration with the highest validation accuracy (86.52%) trained for 3 epochs, while the worst performing configuration (85.04%) trained for 5 epochs. Comparing validation examples where the 3-epoch model succeeded but the 5-epoch model failed reveals that the lower-performing model suffered from **overfitting**, resulting in two distinct patterns of failure:

1. The "Lexical Overlap" Trap (Ignoring Semantic Meaning)

The 5-epoch model frequently misclassified sentences as paraphrases (Label: 1) when they were actually different (Label: 0) simply because they shared a high volume of identical vocabulary.

- **Example:**

- *Sentence 1:* Vivace was founded in 1999 and has raised over \$118 million in three rounds of venture financing.
- *Sentence 2:* During difficult times for technology venture capital, Vivace raised over \$118 million in three rounds of venture financing.

- **Analysis:** The overfit model memorized that heavy keyword matching (e.g., "Vivace", "\$118 million", "three rounds of venture financing") correlates strongly with paraphrases in the training set. It became "lazy" and relied on this word-overlap heuristic, failing to notice the massive semantic difference (one discusses the founding year, the other discusses difficult times for capital). The 3-epoch model retained enough generalization to read the actual semantic meaning of the differing clauses.

2. Structural Rigidity

Conversely, the 5-epoch model failed to recognize actual paraphrases (Label: 1) if the sentence structure was significantly inverted or relied on heavy synonym use.

- **Example:**

- *Sentence 1:* Per-user pricing is \$29 for Workplace Messaging, \$89 for Team Collaboration and \$35 for Collaborative Learning.
- *Sentence 2:* Workplace Messaging is \$29, Workplace Team Collaboration is \$89, and Collaborative Learning is \$35.

- **Analysis:** Here, the exact same information is presented, but the grammar is inverted (Price → Product vs. Product → Price). Because the 5-epoch model memorized rigid syntactic structures from the training data, this structural inversion caused it to predict that they were not paraphrases (Label: 0). The 3-epoch model successfully recognized that despite the grammatical flip, the semantic relationships remained identical.